Project Title: AI resume screening system

Team Name: Biz AI

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1. Problem statement

a. Solution needs

Talent Match Inc. is a social enterprise located in Rwanda whose vision is to bridge the opportunity gap for African youth by facilitating job and internship placements. Currently, their small team spends excessive hours manually reviewing a large pool of applications they receive, which delays placements and has become a significant bottleneck in their workflow. We will deliver an efficient and fair way for Talent Match to connect applicants to opportunities by speeding up the initial candidate screening process without sacrificing quality or fairness. Our solution succeeds if it can match Talent Match's recruiting judgment 90% of the time on past hiring decisions from select test cases. Moreover, the solution will be a success if it produces similar scores when candidate attributes susceptible to bias such as names or gender are changed, and shortlists 120 resumes in under 30 minutes to resolve the operation bottleneck.

b. Requirements

User Requirements:

1. Recruiters shall be able to quickly identify the most relevant candidates from large applicant pools without manual resume review
2. Recruiters shall receive transparent, human-readable explanations for how the candidates were scored.
3. Recruiters shall have candidates grouped into clear job requirements alignment tiers (high, medium, low) to guide their final hiring decisions

System Requirements:

1. The system shall retrieve job descriptions and submitted resumes via Talent Match’s data API.
2. The system must validate input data formats and required fields to prevent incomplete or malformed data from affecting screening
3. The system shall store non-sensitive screening metadata (timestamps, scores, explanations) for auditing and performance analysis.
4. The system shall operate independently from Talent Match's core application with clear separation of concerns.

Functional Requirements:

1. The system must extract required and preferred qualifications from job descriptions.
2. The system must analyze resumes to extract structured candidate attributes (skills, experience, education, certifications, projects).
3. The system must generate custom rubrics for every job before screening candidates.
4. The portal admin shall validate, edit and approve the custom rubric for screening candidates.
5. The system must score and rank candidates based on alignment with role requirements in line with the generated and approved rubric.
6. The system must classify candidates into high, medium, and low alignment tiers.
7. The system must provide clear and concise explanations for each candidate's tier ranking decision.
8. The system must process resumes in PDF and DOCX formats.

Non-Functional Requirements:

1. The system shall screen 120 resumes in no more than 30 minutes.
2. The system shall achieve 90% agreement with Talent's past hiring decisions on a holdout dataset
3. The system shall maintain 0% score variance when candidate attributes associated with bias (name, gender) are changed.
4. The system must provide auditable screening decision logic.
5. The system shall handle variable resume formats without missing critical qualifications.
6. The system must manage API credentials securely and protect candidate data privacy.

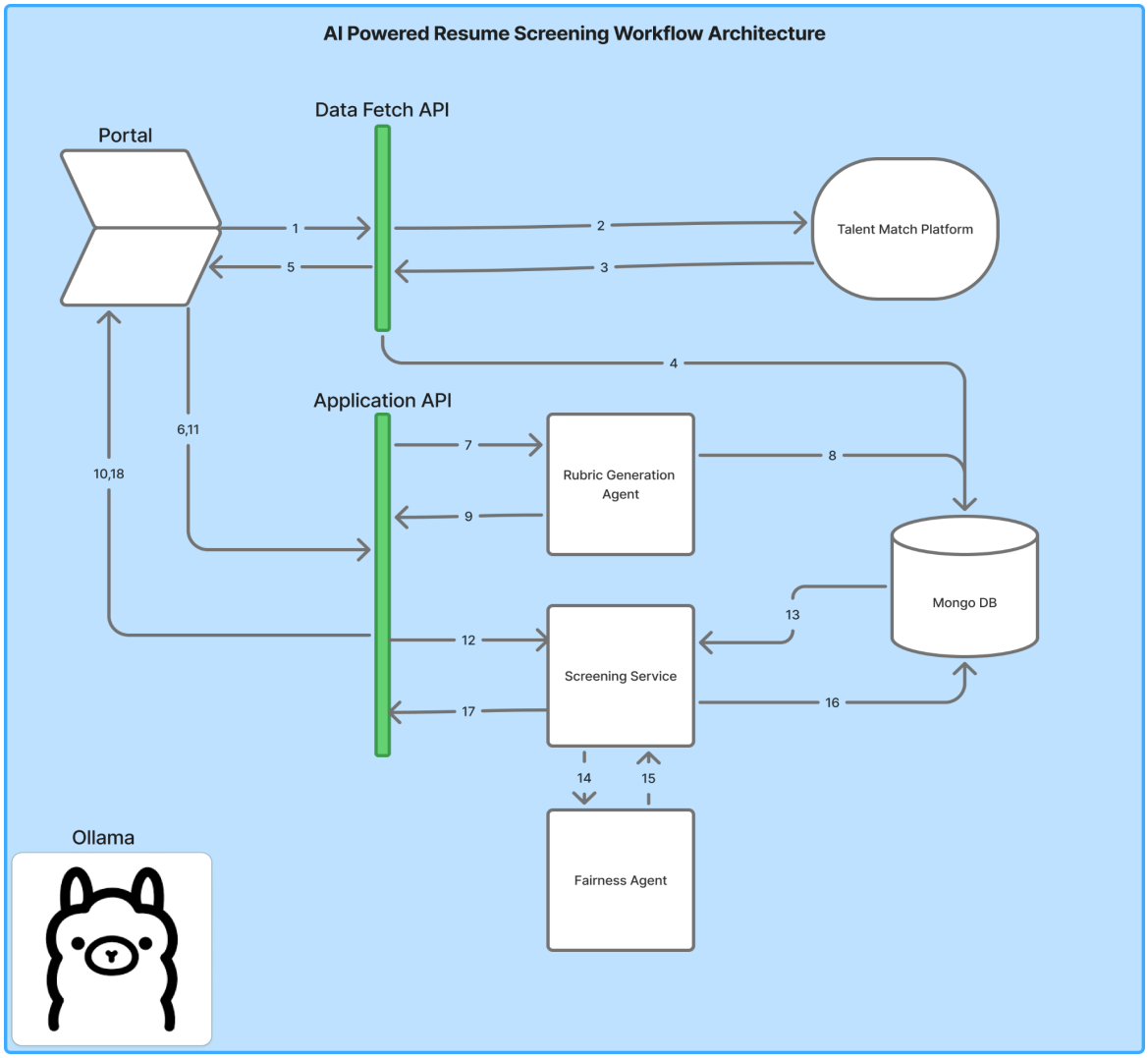
c. Interaction between user and system: automated screening

1. A job posting on the Talent Match platform reaches its application deadline with 120 submitted resumes
2. The recruiter admin logs in to the screening portal and retrieves the job description and all associated resumes via Talent Match APIs
3. The system validates that all retrieved resumes and job data contain required fields (skills, education, experience)
4. The administrator then generates a custom rubric for the job whose applicants are to be screened, and makes any necessary changes to the rubric
5. For each resume, the system extracts candidate attributes and scores each candidate based on the rubric and classifies them into high, medium, and low alignment tiers based on the score.
6. The system presents ranked candidates with their rubric score and explanations to the Talent Match admin portal.
7. The recruiter reviews the top-tier candidates, reads the explanations, and makes final decisions on who to shortlist for interviews
8. The recruiter manually updates candidate statuses (rejected, under review) in the Talent Match platform

d. Services to Be Provided

The system will rely on an automated screening service which will analyze retrieved job descriptions and applicant resumes using natural language processing to evaluate qualification matches without manual intervention. It has a component that generates a detailed rubric that applicants are evaluated on, and this generated rubric must be approved by a Talent Match member before being used in the screening process. The screening component will parse the job

descriptions and applicant information, then use semantic matching to evaluate how well a candidate aligns to a role using the approved rubric and assign scores on different categories. In addition, the system will rely on a fairness agent for bias detection and fairness assurance that identifies and mitigates algorithmic bias to ensure equitable screening across all applicants. It will perform counter factual fairness testing by systematically changing applicant bias prone data and re-run scoring to ensure minimal variance. The system will also use disparate impact analysis ensuring that this ratio is above 80%, while also flagging any detected bias patterns. The system will cluster applicants into three clusters (High, Medium, and Low) reflecting their alignment with the job description, this will be done with a rule based system using the applicants scores. The system will also have an output generation component that outputs the applicants scores, clusters and justification to recruiters for every screening decision, enabling trust. This service will generate summaries for each candidate including highlighting strengths and identifying gaps, with any recommendations. The system will also leverage a validation agent that will detect any hallucinations from the generated final output for correction. Finally, the system will also have an API that will present the results from the screening system to be consumed by the client's platform.



2. Project plan

a. Description and Schedule of Major Activities, Milestones, and Deliverables

From January 26, 2026 to April 24, 2026, this will be implemented in 6 sprints.

Sprint	Activities	Milestones	Deliverables
1.Requirements Analysis	- Analyze Talent Match screening objectives, and current process. - Review authorized API. interfaces and data constraints - Define system architecture and agentic screening workflow. - Identify evaluation metrics and guardrails to manage probabilistic LLM behavior.	- Clear understanding of functional and non-functional requirements - Approved architecture and screening workflow	- High-level system architecture and data flow design - Initial sprint backlog and task breakdown - Initial statement of work (SOW).
2. Initial prototyping	- Implement API-based data retrieval. - Source a screening rubric to enhance LLM objectiveness. - Develop the screening logic. - Implement fairness evaluation.	- API integration validated - Screening logic produces consistent scoring results - Fairness evaluation baseline established	- Working API ingestion module - MVP demo
3. Enhance system robustness	- Thorough testing of the screening system feature to identify and correct any weaknesses. - Incorporating explainability in the screening process.	- System passes defined accuracy threshold - Bias metrics within acceptable range - Explainability output approved	- Improved scoring model - Testing and validation report - Explainability module
4. Enhance system UI/UX and rubric generation automation	- Enhance the system UI to resemble the Talent Match platform. -Implement an agent to auto generate a custom screening rubric for each job.	- UI successfully mirrors Talent Match platform design. - Auto-generated rubric produces usable output	- Implemented frontend interface - Rubric auto-generation module

5. Enhancing system performance and report generation	- Reducing the system response time in rubric generation and resume screening. - Generating reports on screening outcomes. -Allowing user editing of the auto generated rubric.	- System response time reduced to Talent Match satisfaction. - Screening outcome reports validated. - User-editable rubric functionality tested successfully	- Optimized backend performance implementation. - Screening report excel & pdf - Rubric editing interface. -Final SOW.
6. Deployment and user testing	- Package the system for Talent Match on-premise deployment. - Support pilot deployment - Incorporate feedback from user testing as needed.	- Successful local deployment by Talent Match. - Approval from user testing.	- Deployment package (source code, configuration guide) - System documentation.

b. Estimated Effort

Sprint No	Estimated Effort (Weeks)	Weekly hours (1 person)	Total team hours
1	2	12	120
2	2	12	120
3	2	12	120
4	2	12	120
5	2	12	120
6	2	12	120
Total	**12**	**144**	**720**

c. Resource Needs and Responsibility

1. Computing Resources

The project requires local development machines, and containerized runtime support for testing and deployment. These will be supplied by the project team members (local machines).

2. Software Tools and Technologies

The project uses FastAPI for backend development, the Llama 3:latest for LLM-based resume screening and explanation generation, MongoDB for storing non-sensitive screening metadata, and Docker for containerization. These are all open source.

3. Data and Platform Access

The project requires authorized access to Talent Match APIs, including job descriptions, internship postings, resume submissions, status update, and login endpoints.This has been supplied by the Talent Match platform.

4. Plan for Success

a. Criteria for Measuring Outcomes

The success of the project will be measured using objective, repeatable, and team-controlled evaluation methods. All verification will be based on system outputs, automated tests, metrics, and structured demonstrations conducted by the project team.

Outcome Verification Measures

1. Screening Accuracy (Recruiter Alignment Metric)

This will be measured using the percentage agreement between the system's candidate rankings and historical hiring decisions on at least 10 past job applications with a target of 90% agreement.

2. Fairness and Bias Neutralization

This will be measured using score variance when bias-prone attributes like name, and gender are altered (Counterfactual Fairness) targeting 0% variance. The system will rely on the Disparate Impact Ratio across screened candidates with a target of ≥ 0.8. It will also measure Demographic parity which is the prediction accuracy across different demographic groups with a target of equal accuracy despite the demographic group.

3. Performance and Throughput

The system will have a target computation time of at most 5 seconds per resume. This will be assessed by measuring execution time using system timestamps captured during test runs.

4. Explainability and Transparency

The system will measure the structuredness, and human-readability of explanation on candidate's ranking and tier assignment with a target of 100%, and this will be verified by performing automated checks validating the explanation generated for each candidate. The team will also manually inspect at least 15 jobs screening results to ensure completeness and clarity.

5. Milestone Verification

Each sprint will be concluded with a client (contact: allen@talentmatch.rw) and advisor (gokeyo@andrew.cmu.edu) demonstration showing completion of planned functionality. All documents and artifacts will be archived in the project repository for reproducibility and review.

b. Risk Identification and Handling

The team has identified critical risks across organizational, technological, and dependency dimensions that could impact successful project delivery.

Organizational Risks

Risk	Impact	Mitigation Strategy
Team Skill Gaps	Medium - Slower development and potential rework	Conduct skills assessment and allocate the first two weeks for upskilling as needed. Start with simple implementations to test logic and expand later.

Uneven Workload Distribution	Medium - Missed deadlines and poor quality	Establish clear role assignments in Sprint 1. Fair task distribution via sprint planning. Weekly stand-ups to surface issues. Rotate challenging tasks.
Client Communication Clarity	High - Scope drift and frequent rework	Requirements workshop in Sprint 1 with written sign-off. Create sample outputs in Sprint 2 for validation. Weekly then bi-weekly sync meetings.

Technological Risks

Risk	**Impact**	**Mitigation Strategy**
LLM service reliability and performance	High - Development delays and inconsistent screening quality.	Implement error handling and retry logic. Design modular architecture allowing provider switching. Monitor API performance and maintain rule-based fallback mode.
Variable Resume Formats	Medium-High - Missed qualifications and inaccurate rankings	Use proven parsing libraries validated with actual samples in Sprint 1. Build a multi-strategy extraction pipeline. Prioritize critical fields. Flag low-confidence cases.
Platform Integration Complexity	High - Missed deadlines or platform disruption	Early technical meeting with client team. Request staging environment by Sprint 1 end. Design standalone microservice. Implement API mocking.
Missing Job details data from Client API	High - Inaccurate screening results.	Implement automatic completeness checks on required JD fields (skills, responsibilities, experience, qualifications) before screening starts. Block screening if critical fields are missing and alert the user.

Risk Management Process: The team will review this risk register at the end of each sprint during retrospectives. New risks will be added as they emerge, mitigation effectiveness will be evaluated, and high-impact risks will be escalated to the faculty advisor and client proactively.